AdS/CFT and Holographic Duality: Extended Graduate Program

Gustavo Valdivia-Mera

Department of Physics and Texas Center for Superconductivity, University of Houston, Houston, Texas 77204, USA

E-mail: gvaldiviamera@uh.edu

Abstract: This comprehensive 54-week program develops the theoretical foundations of AdS/CFT from the essential background normally acquired in the first-year graduate physics core in the United States, together with graduate-level general relativity. The assumed core consists of standard graduate quantum mechanics, classical mechanics, electrodynamics, statistical mechanics, and the usual mathematical methods of theoretical physics; graduate quantum field theory is not assumed. Nor is prior conformal field theory, large-N gauge theory, supersymmetry, $\mathcal{N} = 4$ super-Yang–Mills theory, string theory, D-brane physics, or holography required. The program therefore builds quantum field theory, renormalization and the renormalization group, conformal field theory, large-N structure, supersymmetry and $\mathcal{N} = 4$ SYM, perturbative string theory and worldsheet consistency, Type-IIB superstrings, D-branes, the D3-brane decoupling construction, the holographic dictionary, holographic renormalization, and canonical applications with substantially greater depth than is strictly necessary for a first introduction to AdS/CFT. Its purpose is to provide a broad derivational foundation from which the correspondence and its standard observables emerge as part of a continuous theoretical development.

Contents

1 Part I: Quantum Field Theory Foundations **2**

1.1 Week 1 — Classical field theory: action, covariance, and Noether symmetry 2

1.2 Week 2 — Free scalar quantum field theory 3

1.3 Week 3 — Interactions, Wick expansion, and the S-matrix 3

1.4 Week 4 — Path integrals and generating functionals 4

1.5 Week 5 — Relativistic fermions: Dirac, Weyl, Majorana, and Grassmann integration 4

1.6 Week 6 — Yang–Mills theory and Wilson operators 5

1.7 Week 7 — Gauge fixing, Faddeev–Popov ghosts, and BRST 5

1.8 Week 8 — Ultraviolet divergences, renormalization, and the quantum effective action 6

1.9 Week 9 — Wilsonian RG, fixed points, and effective field theory 6

1.10 Week 10 — Callan–Symanzik equations, running couplings, and asymptotic freedom 7

1.11 Week 11 — Ward identities, composite operators, and the OPE 7

1.12 Week 12 — Chiral and gauge anomalies 8

1.13 Week 13 — Large N, planar topology, and the gauge/string expansion 8

2 Part II: String-Theory Foundations **9**

2.1 Week 14 — Nambu–Goto, Polyakov, and worldsheet gauge symmetries 9

2.2 Week 15 — Conformal gauge, mode expansions, and classical Virasoro constraints 9

2.3 Week 16 — Light-cone quantization and the bosonic string spectrum 10

2.4 Week 17 — Worldsheet CFT I: OPE, stress tensor, and Ward identities 10

2.5 Week 18 — Worldsheet CFT II: radial quantization, Virasoro, and state–operator map 11

2.6 Week 19 — Polyakov path integral, *bc* ghosts, and the Weyl anomaly 11

2.7 Week 20 — BRST cohomology, physical states, and vertex operators 12

2.8 Week 21 — Tree-level string amplitudes 12

2.9 Week 22 — One-loop torus, modular invariance, and genus expansion 13

2.10 Week 23 — Sigma models, background fields, beta functions, and spacetime equations 13

2.11 Week 24 — Closed-string compactification and T-duality 14

2.12 Week 25 — Open-string T-duality and the emergence of D-branes 14

2.13 Week 26 — Bosonic D-brane dynamics: Chan–Paton, $U(N)$, DBI, and Yang–Mills 15

2.14 Week 27 — RNS worldsheet supersymmetry and NS/R sectors 15

2.15 Week 28 — GSO projection, Type IIA/IIB, and Ramond–Ramond fields 16

2.16 Week 29 — Type-II supergravity and the IIB five-form sector 16

2.17 Week 30 — BPS D-branes, RR charge, DBI+WZ action, and Dp supergravity solutions 17

3 Part III: Spacetime CFT, Supersymmetry, and $\mathcal{N} = 4$ SYM 18
3.1 Week 31 — Spacetime conformal symmetry and conformal kinematics 18
3.2 Week 32 — Radial quantization, state–operator correspondence, unitarity, and OPE 18
3.3 Week 33 — Conserved currents, stress tensor, conformal Ward identities, and anomaly 19
3.4 Week 34 — Supersymmetry algebra, representations, and BPS shortening 19
3.5 Week 35 — Superspace, chiral superfields, and the Wess–Zumino model 20
3.6 Week 36 — Supersymmetric gauge theories 20
3.7 Week 37 — Extended supersymmetry and dimensional reduction toward $\mathcal{N} = 4$ 21
3.8 Week 38 — The $\mathcal{N} = 4$ SYM action, $SU(4)_R$, conformality, and moduli space 21
3.9 Week 39 — Superconformal symmetry, $PSU(2, 2|4)$, and protected operators 22
3.10 Week 40 — Single-trace operators, planar $\mathcal{N} = 4$, and the roles of N and λ 22

4 Part IV: AdS/CFT and Holography 23
4.1 Week 41 — The D3-brane system before the near-horizon limit 23
4.2 Week 42 — Near-horizon decoupling and the emergence of $\mathrm{AdS}_5 \times S^5$ 24
4.3 Week 43 — Symmetry, UV/IR, parameters, and approximation regimes 25
4.4 Week 44 — Field–operator map and mass–dimension relations 26
4.5 Week 45 — GKPW: sources, boundary conditions, and the regulated scalar action 27
4.6 Week 46 — Scalar holographic renormalization and renormalized correlators 28
4.7 Week 47 — Higher-point correlators and Witten diagrams 28
4.8 Week 48 — Metric holographic renormalization, Fefferman–Graham expansion, and $\langle T_{\mu\nu} \rangle$ 29
4.9 Week 49 — Kaluza–Klein/operator matching and protected precision tests 30
4.10 Week 50 — Holographic RG flows and domain walls 31
4.11 Week 51 — Wilson loops and fundamental-string minimal surfaces 32
4.12 Week 52 — Finite temperature: AdS black branes and gauge-theory thermodynamics 33
4.13 Week 53 — Lorentzian holography, retarded functions, Kubo formulae, and transport 34
4.14 Week 54 — Holographic entanglement entropy and synthesis 35

5 References and online resources 36
5.1 Quantum field theory 36
5.2 String theory 36
5.3 Conformal field theory and supersymmetry 37
5.4 AdS/CFT and holography 37

1 Part I: Quantum Field Theory Foundations

Part I develops the QFT structures needed later: classical fields and symmetries, free and interacting quantum fields, path integrals and fermions, Yang–Mills theory and BRST, renormalization and RG, the OPE, anomalies, and the large-N expansion.

The extended route deliberately assigns more formal derivations than the shorter routes. A provisional planning envelope is 12–20 focused hours per week, to be calibrated from completed exercises and the actual pace rather than taken as a guarantee; foundational weeks with several independent calculations may require extra time or be extended without sacrificing prerequisites. Each outcome describes a capability to demonstrate, not a competence assumed to have been acquired.

1.1 Week 1 — Classical field theory: action, covariance, and Noether symmetry

Step	Action	Exact task and source
1	READ	Tong QFT Secs. 1.1–1.3 only. [PDF]
2	DERIVE	From the action, derive the Euler–Lagrange field equations; derive Noether currents for an internal symmetry and for translations; construct the canonical stress tensor and the Lorentz current.
3	DEEPEN	Weinberg I Secs. 7.2–7.4: Lagrangian formalism, global symmetries/Noether generators, energy–momentum tensor and Lorentz invariance.
4	LECTURE	MIT 8.323 Lectures 1–2, after Steps 1–3. [LECTURES]
5	SOLVE	Tong QFT Sheet 1, Problems 2, 3, 5 and 6. [PROBLEMS]
6	OUTCOME	Starting only from a local Lorentz-invariant Lagrangian, reconstruct the field equations, Noether current and stress tensor without consulting notes.

1.2 Week 2 — Free scalar quantum field theory

Step	Action	Exact task and source
1	READ	Tong QFT Secs. 2.1–2.7: canonical quantization, real/complex scalar fields, vacuum/particles, causality and propagators. [PDF]
2	DERIVE	Quantize the real scalar field; derive the mode expansion and commutator algebra; construct the complex scalar charge; derive the Pauli–Jordan commutator and Feynman propagator; verify spacelike microcausality.
3	DEEPEN	Weinberg I Secs. 5.1–5.2, emphasizing how Lorentz covariance, cluster decomposition and causality fix the scalar field/antiparticle structure.
4	LECTURE	MIT 8.323 Lectures 4–6. [LECTURES]
5	SOLVE	Tong QFT Sheet 2, Problems 1–5 and 8. [PROBLEMS]
6	OUTCOME	Derive the free scalar Hilbert space and propagator from canonical commutators and explain precisely where microcausality enters.

1.3 Week 3 — Interactions, Wick expansion, and the S-matrix

Step	Action	Exact task and source
1	READ	Tong QFT Ch. 3 through Sec. 3.7: interaction picture, Dyson expansion, Wick theorem, Feynman rules, cross sections and reduction formula. [PDF]
2	DERIVE	Derive Dyson's formula and Wick's theorem for a scalar interaction; obtain the diagrammatic rules; amputate a connected Green function and recover an on-shell scattering amplitude.
3	DEEPEN	Weinberg I Secs. 3.5 and 6.1–6.4, followed by the LSZ-reduction part of Sec. 10.3.
4	LECTURE	MIT 8.323 Lecture 7 only. The path-integral-dependent Lectures 8–12 are reserved for Week 4. [LECTURES]
5	SOLVE	Tong QFT Sheet 4, Problems 1–2. In addition, derive one connected $2 \to 2$ scalar amplitude from the Dyson expansion through its symmetry/combinatoric factor. [PROBLEMS]
6	OUTCOME	Move from an interaction Lagrangian to a connected amplitude, including symmetry factors and external-state normalization, without treating LSZ reduction as a black box.

1.4 Week 4 — Path integrals and generating functionals

Step	Action	Exact task and source
1	READ	Weinberg I Secs. 9.1–9.4: general path integral, transition to the S-matrix, Lagrangian form and path-integral derivation of Feynman rules.
2	DERIVE	Derive the Gaussian functional integral and $Z_0[J]$ for a free scalar; obtain time-ordered correlators by functional differentiation; expand an interacting $Z[J]$ and recover the Week-3 Feynman rules.
3	LECTURE	MIT 8.323 Lectures 8–10 for the path-integral construction, followed by Lectures 11–12 to connect the path-integral construction with the perturbative expansion. [LECTURES]
4	SOLVE	Calculate $Z_0[J]$ completely for a free scalar and obtain its two-point function by two functional derivatives; then expand a $\lambda\phi^4$ generating functional to first order and recover the connected four-point contribution including contractions.
5	OUTCOME	Translate cleanly between canonical perturbation theory and the functional-integral/generating-functional language.

1.5 Week 5 — Relativistic fermions: Dirac, Weyl, Majorana, and Grassmann integration

Step	Action	Exact task and source
1	READ	Tong QFT Chs. 4–5, with particular attention to Secs. 4.1–4.7 and 5.2–5.7. [PDF]
2	DERIVE	Construct the $(1/2, 0)$ and $(0, 1/2)$ Lorentz representations; derive the Dirac and Weyl equations; impose the Majorana condition; quantize the fermion field and derive its propagator.
3	DEEPEN	Weinberg I Secs. 5.4–5.5 and 9.5; use Sec. 9.5 specifically for Berezin integration and fermionic path integrals.
4	LECTURE	MIT 8.323 Lectures 13–19. [LECTURES]
5	SOLVE	Tong QFT Sheet 3, Problems 1–5. Then evaluate a finite-dimensional Grassmann Gaussian and use it to derive the determinant/inverse-kernel structure of the free fermionic functional integral. [PROBLEMS]
6	OUTCOME	Use Dirac/Weyl/Majorana notation and fermionic functional integration without ambiguity in chirality, conjugation or signs.

1.6 Week 6 — Yang–Mills theory and Wilson operators

Step	Action	Exact task and source
1	READ	Weinberg II Secs. 15.1–15.3: gauge invariance, group structure, Yang–Mills action, field equations and conservation laws.
2	DERIVE	Derive the transformations of A_μ and $F_{\mu\nu}$; derive the Yang–Mills equations and Bianchi identity; derive parallel transport and infinitesimal holonomy around a small loop.
3	DEEPEN	Tong Gauge Theory Ch. 2 through the Wilson-line/Wilson-loop construction. [PDF]
4	LECTURE	MIT 8.324 Lectures 1–4. [LECTURES]
5	SOLVE	MIT 8.324 Problem Set 1, Problems 1–3. [PROBLEMS]
6	OUTCOME	Reconstruct non-Abelian gauge theory from the gauge principle and explain why Wilson loops are natural gauge-invariant observables.

1.7 Week 7 — Gauge fixing, Faddeev–Popov ghosts, and BRST

Step	Action	Exact task and source
1	READ	Weinberg II Secs. 15.4–15.7 in order: quantization, Faddeev–Popov–DeWitt, ghosts, BRST.
2	DERIVE	Derive the FP determinant for a covariant gauge; exponentiate it with ghost fields; write the gauge-fixed action; verify BRST invariance and nilpotence explicitly.
3	LECTURE	MIT 8.324 Lectures 5–6. [LECTURES]
4	SOLVE	MIT 8.324 Problem Set 2, Problem 6; then reconstruct the BRST transformations and the physical-state condition of Lecture 6 independently. [PROBLEMS]
5	OUTCOME	Explain what the ghosts remove, why they are not physical particles, and how BRST cohomology organizes the physical state space.

1.8 Week 8 — Ultraviolet divergences, renormalization, and the quantum effective action

Step	Action	Exact task and source
1	READ	Weinberg I Secs. 12.1–12.3, then Weinberg II Secs. 16.1 and 16.4.
2	DERIVE	Compute the superficial degree of divergence in a representative theory; perform one one-loop subtraction; derive $W[J]$ and the Legendre transform $\Gamma[\phi]$; identify 1PI vertices as derivatives of Γ.
3	LECTURE	MIT 8.324 Lectures 7–9, followed by Lecture 19 for composite-operator singularities, the local OPE structure, and locality of counterterms. [LECTURES]
4	SOLVE	MIT 8.324 Problem Set 4, Problem 2: carry the one-loop ϕ^3 calculation in six dimensions through Feynman parametrization, UV analysis, dimensional regularization and renormalization. [PROBLEMS]
5	OUTCOME	State precisely what is regularized, what is renormalized, why counterterms are local, and what the effective action encodes.

1.9 Week 9 — Wilsonian RG, fixed points, and effective field theory

Step	Action	Exact task and source
1	READ	Weinberg I Sec. 12.4, followed by Weinberg II Secs. 18.4–18.5.
2	DERIVE	Integrate out a momentum shell in a simple model; rescale; linearize the flow around a fixed point; classify relevant, marginal and irrelevant couplings; connect irrelevant operators to an EFT expansion.
3	LECTURE	MIT 8.324 Lectures 20–22, then Lecture 23 through the end of Sec. 5.2.2, including fixed points, relevant/marginal/irrelevant directions, and the critical surface. [LECTURES]
4	SOLVE	MIT 8.324 Problem Set 7, Problem 1(a–d): coupling redefinitions, scheme-independent fixed-point data, RG-invariant scale and dimensional transmutation. [PROBLEMS]
5	OUTCOME	Given a fixed point and a matrix of linearized beta functions, identify the critical surface and the UV/IR fate of every perturbation.

1.10 Week 10 — Callan–Symanzik equations, running couplings, and asymptotic freedom

Step	Action	Exact task and source
1	READ	Weinberg II Secs. 18.2–18.3 and 18.6, followed by Sec. 18.7 only through the beta-function/asymptotic-freedom calculation.
2	DERIVE	Derive a Callan–Symanzik equation; solve a one-coupling RG equation; relate a logarithmic divergence or minimal-subtraction pole to the beta function; reproduce the sign of the Yang–Mills one-loop beta function and its physical meaning.
3	DEEPEN	Tong Gauge Theory Secs. 2.4–2.4.2 for running couplings and asymptotic freedom. [PDF]
4	LECTURE	MIT 8.324 Lecture 23 beginning with Sec. 5.3, followed by Lectures 24–26 for the conventional beta-function/running-coupling formulation. [LECTURES]
5	SOLVE	MIT 8.324 Problem Set 7, Problem 2: compute the beta function of six-dimensional ϕ^3 theory in dimensional regularization/minimal subtraction and establish asymptotic freedom. [PROBLEMS]
6	OUTCOME	Translate without confusion between Wilsonian RG eigen-directions and the conventional running-coupling description.

1.11 Week 11 — Ward identities, composite operators, and the OPE

Step	Action	Exact task and source
1	READ	Weinberg I Sec. 10.4 and Weinberg II Secs. 20.1 and 20.3.
2	DERIVE	Using the functional change-of-variables machinery established in Week 8, derive a Ward identity in the path integral; use the Week-8 composite-operator/OPE locality bridge to compute a free-field short-distance OPE; derive how renormalized composite operators and Wilson coefficients run.
3	SOLVE	Compute the leading short-distance OPE of two free scalar composite operators and one Ward identity with its contact terms. The conformal OPE is rebuilt from symmetry in Part III.
4	OUTCOME	Explain why local operators and their short-distance algebra, rather than only asymptotic particles, are the natural language for CFT and holography.

1.12 Week 12 — Chiral and gauge anomalies

Step	Action	Exact task and source
1	READ	Weinberg II Secs. 22.2–22.4: transformation of the measure, general triangle anomaly and anomaly-free gauge theories.
2	DERIVE	Compute the Fujikawa Jacobian for an infinitesimal chiral rotation; obtain the anomaly equation; reproduce the group-theory tensor controlling a gauge anomaly and the cancellation criterion.
3	DEEPEN	Tong Gauge Theory Secs. 3.2.2–3.2.3, 3.4–3.4.2 and 3.4.4: anomaly in the measure, triangle diagrams, perturbative gauge anomalies, and anomaly cancellation. [PDF]
4	SOLVE	Carry out one explicit Abelian chiral-anomaly calculation and one representation-theory anomaly-cancellation check.
5	OUTCOME	Distinguish an anomalous global symmetry from an inconsistent anomalous gauge symmetry and state exactly what must cancel.

1.13 Week 13 — Large N, planar topology, and the gauge/string expansion

Step	Action	Exact task and source
1	READ	Tong Gauge Theory Secs. 6.2.1 and 6.2.3 for double-line topology, connected-correlator scaling and large-N factorization; use only the topology/string analogy of Sec. 6.2.2, without importing the later $AdS_5 \times S^5$ coupling dictionary. [PDF]
2	DERIVE	For a connected vacuum double-line graph, count faces, edges and vertices and derive $N^{F-E+V} = N^{2-2g}$ at fixed $\lambda = g_{\mathrm{YM}}^2 N$. Then treat single-trace insertions as boundaries, derive the corresponding connected-correlator scaling, and obtain large-N factorization for normalized single-trace operators.
3	DEEPEN	Use Ammon–Erdmenger Sec. 1.7.5 only as a convention check for the 't Hooft limit and topological counting; postpone its explicit holographic identification to Part IV.
4	LECTURE	MIT 8.821 Lectures 7–9 selectively as a preview of the large-N/string-topology analogy; stop before using the specific $\mathcal{N} = 4/AdS_5 \times S^5$ correspondence as an input. [LECTURES]
5	SOLVE	Draw one planar and one genus-one double-line vacuum graph and compute their N scaling; repeat for a connected correlator of normalized single-trace operators.
6	OUTCOME	Explain why the $1/N$ expansion is topological while keeping it distinct from the later worldsheet derivation of the actual string genus coupling.

2 Part II: String-Theory Foundations

Part II proceeds from the classical Polyakov string and its constraints to light-cone quantization and the bosonic spectrum, then to worldsheet CFT, covariant ghost/BRST quantization, amplitudes, target-space dynamics, T-duality, D-branes, the RNS superstring, Type-II spectra, and Type-IIB supergravity.

2.1 Week 14 — Nambu–Goto, Polyakov, and worldsheet gauge symmetries

Step	Action	Exact task and source
1	READ	Tong String Theory Ch. 1 through the Polyakov action, Weyl invariance and classical equations. [PDF]
2	DERIVE	Vary the Nambu–Goto and Polyakov actions; eliminate the auxiliary worldsheet metric to show classical equivalence; derive worldsheet diffeomorphism and Weyl symmetries; derive $T_{ab} = 0$ as the metric equation.
3	DEEPEN	Polchinski I Secs. 1.1–1.2.
4	SOLVE	Tong String Sheet 1, Problems 2(a,b), 3 and 4. [PROBLEMS]
5	OUTCOME	Starting from the geometric string action, reproduce the Polyakov formulation and enumerate its local redundancies exactly.

2.2 Week 15 — Conformal gauge, mode expansions, and classical Virasoro constraints

Step	Action	Exact task and source
1	READ	Tong String Theory Ch. 1, Secs. 1.3.2–1.4.1, for conformal gauge, the closed-string mode expansion, and classical constraints. [PDF]
2	DERIVE	Fix conformal gauge; derive the closed-string solution from periodicity, and derive the open-string mode expansion directly from the Neumann/Dirichlet boundary conditions established in Week 14; obtain $T_{++} = T_{--} = 0$ and the classical Virasoro-mode constraints without quantizing.
3	DEEPEN	Revisit Polchinski I Sec. 1.2 only for the open/closed boundary-condition structure. Reserve Secs. 1.3–1.4 in full for the quantum spectrum of Week 16.
4	SOLVE	Tong String Sheet 1, Problems 5–6. [PROBLEMS]
5	OUTCOME	Write the full open/closed classical mode expansions and constraints without importing quantum conclusions prematurely.

2.3 Week 16 — Light-cone quantization and the bosonic string spectrum

Step	Action	Exact task and source
1	READ	Tong String Theory Ch. 2 for closed-string light-cone/covariant quantization, oscillator algebra, level matching, critical dimension, and the massless graviton/B/Φ sector. [PDF]
2	DERIVE	Fix light-cone gauge; solve the constraints for X^-; quantize the transverse oscillators; compute the normal-ordering constant; derive mass formulae and level matching; identify the tachyon, open-string gauge boson and closed-string graviton/B/Φ states.
3	DEEPEN	Polchinski I Secs. 1.3–1.4 as the explicit open/closed spectrum comparison, including the open-string quantization needed before Tong Sheet 1 Problem 9.
4	SOLVE	Tong String Sheet 1, Problems 8–9. [PROBLEMS]
5	OUTCOME	Derive the physical bosonic spectrum and distinguish light-cone gauge conveniences from gauge-invariant physical content.

2.4 Week 17 — Worldsheet CFT I: OPE, stress tensor, and Ward identities

Step	Action	Exact task and source
1	READ	Polchinski I Secs. 2.1–2.5.
2	DERIVE	Derive the free-boson OPE; construct $T(z)$; compute the $T(z)X(w)$ and $T(z)V(w)$ OPEs; derive conformal Ward identities and the free-boson central charge.
3	DEEPEN	Tong String Theory Ch. 4 through the OPE, stress-tensor and central-charge construction. [PDF]
4	SOLVE	Tong String Sheet 2, Problems 1–3 and 5. [PROBLEMS]
5	OUTCOME	Compute conformal weights and central charges directly from OPEs and use Ward identities rather than asserted transformation rules.

2.5 Week 18 — Worldsheet CFT II: radial quantization, Virasoro, and state–operator map

Step	Action	Exact task and source
1	READ	Polchinski I Secs. 2.6–2.9.
2	DERIVE	Map the cylinder to the plane; construct radial quantization; derive the Virasoro algebra from the TT OPE; establish the state–operator map, highest-weight conditions and oscillator/operator correspondence.
3	DEEPEN	Tong String Theory Ch. 4, radial-quantization, Virasoro and state–operator subsections. [PDF]
4	SOLVE	Tong String Sheet 2, Problems 4, 7 and 8. [PROBLEMS]
5	OUTCOME	Move fluently among local operators on the plane, states on the cylinder and Virasoro representations. The generic bc CFT has already appeared mathematically; its identification as the Faddeev–Popov ghost system of the Polyakov path integral is established in Week 19.

2.6 Week 19 — Polyakov path integral, bc ghosts, and the Weyl anomaly

Step	Action	Exact task and source
1	READ	Polchinski I Secs. 3.1–3.6; leave curved target-space backgrounds of Sec. 3.7 for Week 23.
2	DERIVE	Gauge-fix Diff×Weyl in the Polyakov path integral; derive the FP ghost system; compute $c_{bc} = -26$; combine ghost and matter central charges and recover $D = 26$ from anomaly cancellation.
3	DEEPEN	Tong String Theory Ch. 5 through the ghost CFT and critical-dimension construction. [PDF]
4	SOLVE	Tong String Sheet 2, Problem 6; independently rederive the bc conformal weights and central charge. [PROBLEMS]
5	OUTCOME	Explain the critical dimension as quantum consistency of the worldsheet gauge symmetry, not as an unexplained numerical output of light-cone quantization.

2.7 Week 20 — BRST cohomology, physical states, and vertex operators

Step	Action	Exact task and source
1	READ	Polchinski I Secs. 4.1–4.3.
2	DERIVE	Construct the BRST current and charge; verify nilpotence when the total central charge vanishes; characterize physical states by BRST cohomology; construct the first BRST-closed unintegrated vertex representatives and derive their cohomological gauge equivalences.
3	DEEPEN	Tong String Theory Ch. 5 as a non-BRST comparison for physical-state and matter vertex-operator conformal-weight intuition; Tong explicitly does not develop BRST quantization there. [PDF]
4	SOLVE	Check BRST closure for the first massless open- and closed-string vertex operators and derive their gauge redundancies as BRST-exact shifts.
5	OUTCOME	Recover the physical-state constraints covariantly and explain why BRST cohomology is the gauge-invariant Hilbert space.

2.8 Week 21 — Tree-level string amplitudes

Step	Action	Exact task and source
1	READ	Polchinski I Secs. 5.1–5.3, the BRST-invariance/integrated-versus-unintegrated insertion part of Sec. 5.4, and Secs. 6.1–6.6: moduli/measure ingredients, disk/sphere tree amplitudes, Chan–Paton factors and open/closed tree-level processes.
2	DERIVE	Explain why fixed/unintegrated vertices carry ghost insertions while integrated matter vertices are BRST invariant up to total derivatives; then fix residual conformal Killing symmetry, derive the Koba–Nielsen structure, compute one four-point tree amplitude and identify its pole factorization into the string spectrum.
3	DEEPEN	Tong String Theory Ch. 6 through the Veneziano and Virasoro–Shapiro amplitudes. [PDF]
4	SOLVE	Tong String Sheet 3, Problems 2 and 4. [PROBLEMS]
5	OUTCOME	Set up and evaluate a tree-level worldsheet amplitude with correct gauge fixing, moduli and vertex-operator normalization structure.

2.9 Week 22 — One-loop torus, modular invariance, and genus expansion

Step	Action	Exact task and source
1	READ	Polchinski I Secs. 7.1–7.3.
2	DERIVE	Describe torus moduli and the modular group; derive the one-loop partition-function measure structurally; verify modular invariance of the basic free-boson building blocks; derive the Euler-characteristic/genus weighting of perturbative closed strings.
3	DEEPEN	Tong String Theory Ch. 6, torus/modular and one-loop partition-function material. [PDF]
4	SOLVE	Show explicitly how the dilaton coupling to the worldsheet Euler characteristic produces the genus factor, and verify invariance of the torus free-boson partition function under T and S transformations.
5	OUTCOME	Connect the worldsheet topological expansion to the large-N topology of Week 13 without identifying their couplings before the AdS/CFT dictionary is established.

2.10 Week 23 — Sigma models, background fields, beta functions, and spacetime equations

Step	Action	Exact task and source
1	READ	Polchinski I Sec. 3.7.
2	DERIVE	Write the nonlinear sigma model with $G_{\mu\nu}$, $B_{\mu\nu}$ and Φ; derive the leading one-loop/linearized structure of the metric beta function and understand its covariant completion $\beta^G_{\mu\nu} = \alpha' R_{\mu\nu} + \cdots$; show how vanishing worldsheet beta functions become target-space field equations.
3	DEEPEN	Tong String Theory Ch. 7 through the background-field beta functions, B field, dilaton and low-energy effective action. [PDF]
4	SOLVE	Tong String Sheet 4, Problems 1–2. [PROBLEMS]
5	OUTCOME	Explain precisely how quantum Weyl invariance constrains target-space dynamics and why the graviton obeys Einstein dynamics at leading low energy.

2.11 Week 24 — Closed-string compactification and T-duality

Step	Action	Exact task and source
1	READ	Polchinski I Secs. 8.1–8.4.
2	DERIVE	Quantize a compact boson on S^1; derive momentum and winding contributions; obtain the mass formula and its invariance under $R \leftrightarrow \alpha'/R$ with momentum–winding exchange; identify the self-dual enhancement.
3	DEEPEN	Tong String Theory Ch. 8 through Sec. 8.3.1 only: closed-string compactification and T-duality. Reserve open-string T-duality for Week 25 and Type-II T-duality for the superstring stage. [PDF]
4	SOLVE	Derive p_L, p_R, level matching and the additional massless states at the self-dual radius directly from the worldsheet spectrum.
5	OUTCOME	Treat T-duality as an exact equivalence of the quantum string spectrum, not merely a formal coordinate change.

2.12 Week 25 — Open-string T-duality and the emergence of D-branes

Step	Action	Exact task and source
1	READ	Polchinski I Secs. 8.6–8.7 through the T-dual open-string boundary conditions and D-brane interpretation.
2	DERIVE	T-dualize a Neumann direction and obtain Dirichlet boundary conditions; show how Wilson lines map to brane positions; derive the spectrum of open strings stretched between separated branes.
3	DEEPEN	Tong String Theory Sec. 8.3.2 plus the selected Ch. 3 boundary-condition/stretched-string material needed to interpret the D-brane; omit Sec. 3.1.4 and reserve the later brane-dynamics/multiple-brane material for Week 26. [PDF]
4	SOLVE	Tong String Sheet 1, Problem 10. [PROBLEMS]
5	OUTCOME	Derive D-branes from open-string T-duality before treating them as dynamical nonperturbative objects.

2.13 Week 26 — Bosonic D-brane dynamics: Chan–Paton, $U(N)$, DBI, and Yang–Mills

Step	Action	Exact task and source
1	READ	Polchinski I Sec. 8.7, now focusing on coincident branes, matrix coordinates, tension and the Born–Infeld action.
2	DERIVE	Derive $U(N)$ Chan–Paton gauge symmetry for N coincident branes; expand DBI to quadratic order and recover worldvolume Yang–Mills; relate adjoint scalar matrices to transverse brane fluctuations.
3	DEEPEN	Tong String Theory selected multiple-brane material from Ch. 3 and Ch. 7 through Sec. 7.7, stopping before Sec. 7.7.1 (Type-II/BPS D-branes). [PDF]
4	SOLVE	Tong String Sheet 4, Problem 5(a); then expand the abelian DBI determinant through F^2 and scalar-gradient terms and identify the worldvolume Yang–Mills coupling. [PROBLEMS]
5	OUTCOME	Explain why coincident D-branes produce non-Abelian gauge theory and keep this bosonic DBI/Yang–Mills construction, including its couplings to $G_{\mu\nu}$, $B_{\mu\nu}$ and Φ, distinct from RR charge, which appears only after the superstring spectrum is built.

2.14 Week 27 — RNS worldsheet supersymmetry and NS/R sectors

Step	Action	Exact task and source
1	READ	Polchinski II Secs. 10.1–10.4.
2	DERIVE	Construct the RNS action and superconformal currents; impose NS and R boundary conditions; quantize the worldsheet fermions; derive the super-Virasoro algebra and zero-point shifts; construct the superghost sector at the level needed for physical states.
3	SOLVE	Compute the NS/R zero-point shifts, the lowest unprojected NS and R Fock-space states, and the action of fermion zero modes on the Ramond ground state; record the relevant ghost-ground-state data without yet imposing the full physical-state/GSO projection.
4	OUTCOME	Explain how the unprojected NS and R worldsheet sectors furnish candidate spacetime bosonic and fermionic states; reserve the physical-state constraints, GSO projection, and final Type-II spectrum for Week 28.

2.15 Week 28 — GSO projection, Type IIA/IIB, and Ramond–Ramond fields

Step	Action	Exact task and source
1	READ	Polchinski II Secs. 10.5–10.7.
2	DERIVE	Implement the GSO projection; construct the massless NS–NS, NS–R, R–NS and R–R sectors; derive the chirality distinction between IIA and IIB and identify the RR form fields.
3	SOLVE	Build the complete massless Type-II field-content table from left/right sectors and verify level matching, chirality and spacetime statistics.
4	OUTCOME	State the complete massless IIA/IIB bosonic content and explain why Type IIB, not Type IIA, contains an RR four-form potential C_4 whose five-form field strength F_5 is self-dual and can support the later $\mathrm{AdS}_5 \times S^5$ solution.

2.16 Week 29 — Type-II supergravity and the IIB five-form sector

Step	Action	Exact task and source
1	READ	Polchinski II Sec. 12.1: low-energy supergravity of the Type-II theories.
2	DEEPEN	Ammon–Erdmenger Sec. 4.2.3, especially the Type-IIB supergravity subsection and its bosonic field-content table.
3	DERIVE	Write the bosonic IIB field content in one convention; distinguish NS–NS and RR field strengths; state the self-duality condition on F_5 and explain why the usual covariant action must be supplemented by this condition.
4	SOLVE	Check the form ranks, gauge invariances and mass dimensions of the IIB bosonic fields and verify the stress-energy structure that allows a five-form-supported product geometry.
5	OUTCOME	Have a working low-energy IIB supergravity framework before using D3-brane geometries or five-form flux quantization.

2.17 Week 30 — BPS D-branes, RR charge, DBI+WZ action, and Dp supergravity solutions

Step	Action	Exact task and source
1	READ	Polchinski II Sec. 13.1, then the selected part of Sec. 13.2 deriving the half-BPS D-brane supersymmetry projector and RR charge algebra, followed by Sec. 13.3 through charge/no-force/DBI+WZ and the tension recursion. Omit Sec. 13.4 from the core route.
2	DEEPEN	Ammon–Erdmenger Sec. 4.4.1 from the RR/Wess–Zumino coupling onward, then Sec. 4.4.2 through the general Dp harmonic-function and flux-charge construction, stopping before the D3-specific specialization.
3	DERIVE	Track Type-II T-duality on chirality and RR forms; show why D-branes carry RR charge; write DBI plus Wess–Zumino/Chern–Simons couplings; identify the preserved half of spacetime supersymmetry and derive the corresponding tension/charge no-force cancellation; derive the harmonic-function equation of the Dp geometry. The algebraic BPS bound and shortening are derived in Week 34.
4	SOLVE	Derive the recursion of D-brane tension/charge under T-duality and the transverse-space scaling $H_p(r) = 1 + \text{const.}/r^{7-p}$ for a localized Dp source.
5	OUTCOME	Move consistently from perturbative open strings to BPS RR-charged branes and their low-energy supergravity geometries. The D3 system is now ready for Part IV.

3 Part III: Spacetime CFT, Supersymmetry, and $\mathcal{N} = 4$ SYM

This part converts the RG/operator language of Part I into the representation-theoretic language needed on the CFT side of the duality, then constructs the specific superconformal gauge theory that appears in AdS_5/CFT_4.

3.1 Week 31 — Spacetime conformal symmetry and conformal kinematics

Step	Action	Exact task and source
1	READ	Rychkov Lectures 1–2: RG origin, conformal group, primaries and kinematics. [PDF]
2	DERIVE	Derive the conformal algebra in $d > 2$; derive scalar-primary transformation laws; fix scalar two- and three-point functions; derive the two independent cross-ratios of a scalar four-point function.
3	DEEPEN	Ammon–Erdmenger Secs. 3.2.1–3.2.2, with only the scalar-correlator normalization material of Sec. 3.2.4 needed to harmonize conventions. Reserve Sec. 3.2.3 for Week 33.
4	OUTCOME	Derive scalar two- and three-point structures from symmetry and identify exactly what remains dynamical at four points.

3.2 Week 32 — Radial quantization, state–operator correspondence, unitarity, and OPE

Step	Action	Exact task and source
1	READ	Rychkov Lecture 3 in full. [PDF]
2	DERIVE	Construct radial quantization; derive the state–operator correspondence; obtain scalar and symmetric-traceless unitarity bounds from positivity of descendant norms; organize the OPE into conformal families.
3	DEEPEN	Simmons–Duffin Sec. 7.3, especially Sec. 7.3.1, for the clean link between saturation of a unitarity bound, null descendants, and conservation. [PDF]
4	SOLVE	Starting from the conformal algebra, compute the norm matrix of first-level descendants and reproduce one unitarity bound; then expand a scalar two-point function in radial quantization and match the first descendant coefficient.
5	OUTCOME	Translate freely among local operators, states on S^{d-1}, primary/descendant modules and OPE data.

3.3 Week 33 — Conserved currents, stress tensor, conformal Ward identities, and anomaly

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Secs. 3.2.3–3.2.6: correlators/OPE, renormalized CFTs, conformal Ward identities and trace anomaly.
2	DEEPEN	Simmons–Duffin Sec. 2.1 for the stress tensor as the response to a background metric and its Ward identity. [PDF]
3	DERIVE	Derive the Ward identity for a conserved current and for $T_{\mu\nu}$; derive $\Delta_J = d - 1$ and $\Delta_T = d$ from conservation/unitarity; relate the trace anomaly to the renormalized generating functional in a curved background.
4	SOLVE	Derive the conformally fixed two-point tensor structure of a conserved current and verify one stress-tensor Ward identity including its contact terms.
5	OUTCOME	Derive the protected dimensions and tensor structures of J_μ and $T_{\mu\nu}$, their Ward identities, and the background-metric source definition of $T_{\mu\nu}$; leave the bulk field/operator and source/response dictionary for Part IV.

3.4 Week 34 — Supersymmetry algebra, representations, and BPS shortening

Step	Action	Exact task and source
1	READ	Tong SUSY Ch. 2 through Sec. 2.4.2; stop before Sec. 2.4.3 on supersymmetry in other dimensions/dimensional reduction. [PDF]
2	DERIVE	Construct the four-dimensional $\mathcal{N} = 1$ super-Poincare algebra and its massless multiplets; then extend the algebra to $\mathcal{N} > 1$, identify the allowed central charges, and derive the BPS bound and shortening from positivity of the extended supercharge anticommutator.
3	DEEPEN	Weinberg III Secs. 25.2 and 25.5 for the general central-charge algebra, BPS bound and shortened massive multiplets.
4	SOLVE	As a brief convention warm-up, use Tong SUSY Sheet 1, Problems 1–3. Then construct one $\mathcal{N} = 1$ massless multiplet directly from helicity-raising supercharges and, for $\mathcal{N} = 2$, diagonalize the central-charge algebra, derive $m \geq \lvert Z \rvert$, and count the long versus saturated states. [PROBLEMS]
5	OUTCOME	Construct a supermultiplet directly from the algebra and distinguish long from shortened/BPS representations.

3.5 Week 35 — Superspace, chiral superfields, and the Wess–Zumino model

Step	Action	Exact task and source
1	READ	Tong SUSY Secs. 3.1–3.2.3 and 3.3.1–3.3.2: superspace, chiral superfields, F/D terms, the Wess–Zumino model, and the first R-symmetry/holomorphy non-renormalization argument. [PDF]
2	DERIVE	Realize supercharges and covariant derivatives on superspace; solve the chirality constraint; derive F- and D-term integrals; expand a general renormalizable chiral action and eliminate auxiliary fields.
3	DEEPEN	Weinberg III Secs. 26.3–26.4 for chiral superfields, superpotentials and the general renormalizable chiral theory.
4	SOLVE	Tong SUSY Sheet 1, Problems 7, 9 and 10: superspace operators, the chiral kinetic/superpotential integrals, and the Wess–Zumino action. [PROBLEMS]
5	OUTCOME	Move between superfield and component formulations and derive the Wess–Zumino action rather than memorizing it.

3.6 Week 36 — Supersymmetric gauge theories

Step	Action	Exact task and source
1	READ	Tong SUSY Secs. 4.1–4.2.2: vector superfields, field strengths, Abelian and non-Abelian Super Yang–Mills, and chiral matter; stop before moduli spaces/extended supersymmetry. [PDF]
2	DERIVE	Construct the vector superfield and W_α; impose Wess–Zumino gauge; derive the SYM component action; couple chiral matter; eliminate F and D fields and derive the scalar potential.
3	DEEPEN	Weinberg III Sec. 27.4 as the main complementary construction of the general renormalizable supersymmetric gauge theory with chiral matter; use Secs. 27.1–27.3 only for convention checks as needed.
4	SOLVE	Tong SUSY Sheet 2, Problems 3–4; then eliminate the F_i and D^A auxiliary fields in a gauge theory with chiral matter and obtain $V = V_F + V_D$ explicitly. [PROBLEMS]
5	OUTCOME	Derive the component content and interactions of an $\mathcal{N} = 1$ supersymmetric gauge theory from superspace.

3.7 Week 37 — Extended supersymmetry and dimensional reduction toward $\mathcal{N} = 4$

Step	Action	Exact task and source
1	READ	Weinberg III Sec. 27.9 selectively, concentrating on its $\mathcal{N} = 4$ gauge-theory material rather than repeating the full $\mathcal{N} = 2$ buildup.
2	READ	Ammon–Erdmenger Sec. 3.3.6, specifically the ten-dimensional $\mathcal{N} = 1$ SYM to four-dimensional $\mathcal{N} = 4$ dimensional-reduction construction; stop before the later properties/vacua discussion reserved for Week 38.
3	DERIVE	Reduce ten-dimensional $\mathcal{N} = 1$ SYM on six flat directions; identify the four-dimensional gauge field, six real adjoint scalars and four Weyl fermions; decompose the ten-dimensional Yang–Mills kinetic term into the four-dimensional gauge kinetic term, scalar kinetic terms and the commutator potential, fixing all signs and numerical factors in the action convention used; identify $\mathrm{Spin}(6) \cong SU(4)_R$ (equivalently $\mathfrak{so}(6) \cong \mathfrak{su}(4)$), with the scalars in the $\mathbf{6}$.
4	SOLVE	Use Ammon–Erdmenger Exercise 3.3.18 for the fermionic reduction and independently complete the bosonic dimensional reduction of the ten-dimensional Yang–Mills kinetic term.
5	OUTCOME	Obtain the $\mathcal{N} = 4$ multiplet from a structural construction rather than a memorized field-content table.

3.8 Week 38 — The $\mathcal{N} = 4$ SYM action, $SU(4)_R$, conformality, and moduli space

Step	Action	Exact task and source
1	READ	Resume Ammon–Erdmenger Sec. 3.3.6 after the dimensional-reduction construction for the four-dimensional $\mathcal{N} = 4$ action, stated properties and vacua.
2	DERIVE	Write the complete four-dimensional action in one normalization; identify the $SU(4)_R \cong \mathrm{Spin}(6)$ representation content; derive the scalar commutator potential and moduli-space condition $[\phi^i, \phi^j] = 0$; show that all dynamical fields transform in the adjoint; record $\tau_{\mathrm{YM}} = \theta/(2\pi) + 4\pi i/g^2_{\mathrm{YM}}$ in the convention used here.
3	DEEPEN	Use the $\mathcal{N} = 4$ tail of Weinberg III Sec. 27.9 for the established vanishing-beta-function/finiteness result and electric–magnetic duality. Distinguish these established statements from the explicit classical/action-level derivations carried out in this week.
4	SOLVE	Check bosonic/fermionic on-shell degree-of-freedom matching and derive generic Coulomb-branch breaking $SU(N) \to U(1)^{N-1}$.
5	OUTCOME	State the action, symmetries, field representations, coupling and moduli space of $\mathcal{N} = 4$ SYM without ambiguity.

3.9 Week 39 — Superconformal symmetry, $PSU(2,2|4)$, and protected operators

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Secs. 3.4.1–3.4.3.
2	DERIVE	Build the bosonic algebra $\mathfrak{so}(4,2) \oplus \mathfrak{su}(4)_R$ inside $\mathfrak{psu}(2,2\|4)$; identify Poincare and conformal supercharges; define superconformal primaries; derive the positivity/annihilation logic for the explicit half-BPS scalar family rather than claiming a complete classification of all shortening conditions.
3	SOLVE	Construct representative half-BPS scalar single-trace operators $\mathrm{Tr}(\phi^{(i_1}\cdots\phi^{i_k)})$ with traces removed; identify the $[0,k,0]$ $SU(4)_R$ family with $\Delta = k$, and identify $k = 2$ specifically as the protected $[0,2,0]$, $\Delta = 2$, $\mathbf{20'}$ half-BPS primary.
4	OUTCOME	Explain why protected dimensions can be compared between weak and strong coupling and why generic operator dimensions cannot.

3.10 Week 40 — Single-trace operators, planar $\mathcal{N} = 4$, and the roles of N and λ

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 3.4.3 for the protected single-trace family; revisit Tong Gauge Theory Sec. 6.2.1 and the connected-scaling/factorization part of Sec. 6.2.3, explicitly skipping Sec. 6.2.2 and any later holographic identification. [PDF]
2	DERIVE	Normalize single-trace operators; derive connected-correlator N scaling and large-N factorization at fixed 't Hooft coupling; distinguish single-trace from multi-trace operators without assigning either a bulk particle interpretation yet; keep $1/N$ as the gauge-theory topological expansion and λ as the parameter controlling planar dynamics. Their precise bulk interpretation is derived only after the D3-brane map and field/operator dictionary.
3	SOLVE	For normalized single-trace operators, derive the N scaling of connected two-, three- and four-point functions and show explicitly how large-N factorization emerges.
4	OUTCOME	State what large N controls on the gauge-theory side, what changing λ changes in planar dynamics, and which bulk interpretation still remains to be derived rather than assumed.

4 Part IV: AdS/CFT and Holography

Part IV begins with the open-string/gauge and closed-string/supergravity descriptions of the D3-brane system. Their common low-energy limit leads to $\mathrm{AdS}_5 \times S^5$ and the AdS/CFT dictionary; the field–operator map and holographic renormalization then provide the framework for correlators, RG flows, Wilson loops, finite-temperature physics, real-time response, and entanglement.

4.1 Week 41 — The D3-brane system before the near-horizon limit

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 5.2.1 and Sec. 5.2.2 through Eqs. (5.11)–(5.13), with a targeted revisit of Sec. 4.4.2 only for the D3 flux-quantization normalization.
2	DERIVE	For N coincident D3-branes, derive the massless open-string worldvolume content and recover four-dimensional $\mathcal{N} = 4$ $U(N)$ SYM. On the closed-string side, write the extremal D3 metric, $H(r) = 1 + L^4/r^4$, constant dilaton and self-dual five-form flux, and derive $L^4 = 4\pi g_s N \alpha'^2$ from flux quantization. Defer the holographic removal of the free center-of-mass $U(1)$ to the combined decoupling argument of Week 42.
3	DEEPEN	Optional normalization cross-check: selected Polchinski II Sec. 13.3 material relating D-brane RR charge and tension to the supergravity source.
4	LECTURE	MIT 8.821 Lectures 13–15: D-branes I–III. [LECTURES]
5	OUTCOME	Describe the same D3 stack in the open-string/gauge and closed-string/supergravity languages before taking any near-horizon limit.

4.2 Week 42 — Near-horizon decoupling and the emergence of $\mathrm{AdS}_5 \times S^5$

Step	Action	Exact task and source
1	READ	Finish Ammon–Erdmenger Sec. 5.2.2 and then Sec. 5.2.3, in that order.
2	READ	Maldacena, *The Large N Limit of Superconformal Field Theories and Supergravity*, introduction and D3-brane decoupling argument. [PAPER]
3	DERIVE	Take $\alpha' \to 0$ with $u = r/\alpha'$ and the Yang–Mills coupling fixed; show that massive open-string modes decouple while the asymptotically flat closed-string sector decouples from the throat; approximate $H(r) \to L^4/r^4$ and obtain $\mathrm{AdS}_5 \times S^5$ with common radius L. Explain the throat redshift: finite gauge-theory energies can correspond to finite proper energies in string units, so the decoupled near-horizon sector contains full Type-IIB string excitations rather than only supergravity zero modes. Separate the free center-of-mass $U(1)$ from the interacting $SU(N)$ sector.
4	LECTURE	MIT 8.821 Lectures 16–17: D-brane geometry, the AdS/CFT conjecture and its immediate consequences. [LECTURES]
5	SOLVE	MIT 8.821 Problem Set 4, Problems 1–3: AdS conformal transformations, global-AdS geodesic motion, and geodesic distance. The IR/UV problem is deferred to Week 43, after the UV/IR map is derived. [PROBLEMS]
6	OUTCOME	Reconstruct how the two low-energy descriptions of one D3 system motivate the AdS/CFT conjecture with full Type-IIB string theory on $\mathrm{AdS}_5 \times S^5$; do not treat the decoupling argument as a mathematical proof or use "near horizon = AdS/CFT" as an unexplained input.

4.3 Week 43 — Symmetry, UV/IR, parameters, and approximation regimes

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 5.1 and the coupling/limit discussion surrounding the D3 construction in Sec. 5.2.
2	DERIVE	Starting from the AdS hyperboloid in $\mathbb{R}^{d,2}$, write the global and Poincare forms of the metric, identify the conformal boundary in each description, and show explicitly how Poincare dilatations act as a radial rescaling. Then match $SO(4,2)$ AdS isometries with four-dimensional conformal symmetry and the S^5 isometry algebra $\mathfrak{so}(6)$ with the CFT $\mathfrak{su}(4)_R$ algebra, with $SU(4)_R \cong$ Spin(6) when spinorial representations matter; derive the radial UV/IR relation from this scaling action. Convert the normalization used in Ammon–Erdmenger to the convention of this program and establish $L^4 = 4\pi g_s N \alpha'^2$, $g_{\mathrm{YM}}^2 = 4\pi g_s$, $\lambda = g_{\mathrm{YM}}^2 N = L^4/\alpha'^2$, $g_s = \lambda/(4\pi N)$, and $\alpha'/L^2 = \lambda^{-1/2}$. Using $2\kappa_{10}^2 = (2\pi)^7 g_s^2 \alpha'^4$, $G_{10} = \kappa_{10}^2/(8\pi)$, and $\mathrm{Vol}(S^5) = \pi^3 L^5$, dimensionally reduce to $G_5 = G_{10}/\mathrm{Vol}(S^5) = \pi L^3/(2N^2)$.
3	DERIVE	Separate the expansion regimes: finite (N, λ) corresponds to full quantum IIB string theory; $N \to \infty$ at fixed λ suppresses string loops and leaves genus-zero string theory while α' corrections remain unsuppressed in general; classical two-derivative supergravity additionally requires $\lambda \gg 1$ and $g_s = \lambda/(4\pi N) \ll 1$.
4	LECTURE	MIT 8.821 Lecture 18 through Sec. 3.1.3 only: UV/IR, symmetry matching, and parameter matching. Stop before Sec. 3.1.4 introduces the field/operator map of Week 44. [LECTURES]
5	SOLVE	MIT 8.821 Problem Set 4, Problems 4–5: the IR/UV connection and holographic degree counting, now after the corresponding UV/IR and parameter derivations. [PROBLEMS]
6	OUTCOME	Given (N, λ), state which bulk description is controlled, which correction parameter is small, and which conclusion does *not* follow from large N alone.

4.4 Week 44 — Field–operator map and mass–dimension relations

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Secs. 5.3.3–5.3.4 as the backbone for the field/operator representation map and scalar near-boundary/Breitenlohner–Freedman analysis; use only the minimal notation/context needed from Secs. 5.3.1–5.3.2, whose detailed KK/protected matching is reserved for Week 49. Use Klebanov–Witten as the primary reference for the two admissible scalar quantizations in the allowed mass window. [PAPER] For the distinction between an indicial root and stability, consult the original Breitenlohner–Freedman positive-energy analysis; use Klebanov–Witten specifically for boundary conditions and the two operator dimensions. [PAPER]
2	DERIVE	Solve the scalar wave equation near the AdS boundary and obtain $m^2L^2 = \Delta(\Delta - d)$ and $\Delta_\pm = d/2 \pm \nu$, $\nu^2 = d^2/4 + m^2L^2$. The reality of ν identifies a candidate threshold; to justify its interpretation as the Breitenlohner–Freedman stability bound, rewrite the Poincare radial problem as a one-dimensional inverse-square Schrödinger operator and use the sharp Hardy inequality to test boundedness from below, stating the allowed energy-conserving boundary conditions. Establish $m^2L^2 \geq -d^2/4$ and the generic alternate-quantization window $-d^2/4 < m^2L^2 < -d^2/4 + 1$; treat the saturated/logarithmic endpoint separately. Apply the analogous asymptotic argument to a vector and identify the conserved-current and stress-tensor sources.
3	LECTURE	MIT 8.821 Lecture 19: mass–dimension relation. [LECTURES]
4	SOLVE	MIT 8.821 Problem Set 5, Problem 1 for vectors; independently show that for $\phi = z^{(d-1)/2}\psi$ the scalar radial equation has inverse-square coefficient $m^2L^2 + (d^2 - 1)/4$ in $-\partial_z^2\psi + V(z)\psi$, and identify where a boundary-flux condition enters. The full global-energy proof is assigned to the original Breitenlohner–Freedman reference, not inferred solely from the indicial equation. [PROBLEMS]
5	OUTCOME	Infer the CFT dimension and tensor character associated with basic bulk scalar, vector and metric fields directly from asymptotics and gauge symmetry.

4.5 Week 45 — GKPW: sources, boundary conditions, and the regulated scalar action

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 5.4.1, the scalar propagator material needed from Sec. 5.4.2, and Sec. 5.4.3 only through the regulated cutoff-surface action; defer systematic counterterms and the renormalized variational problem to Week 46.
2	READ	Gubser–Klebanov–Polyakov and Witten, focusing on the partition/generating-functional relation, boundary data and the semi-classical bulk limit. [PAPER] [PAPER]
3	DERIVE	State first the exact conjectural relation between the CFT generating functional and the full string partition function with specified asymptotic data. Then pass explicitly to its classical-gravity approximation at finite cutoff, $Z_{\mathrm{CFT},\epsilon} \sim e^{-S^{\mathrm{os}}_{\mathrm{reg}}}$. For a Euclidean scalar in AdS_{d+1}, solve the bulk equation with prescribed boundary source, construct the bulk-to-boundary propagator, reduce the on-shell action to $z = \epsilon$, and display the divergent local terms before removing any of them; reserve the finite $e^{-S^{\mathrm{os}}_{\mathrm{ren}}}$ variational statement for Week 46.
4	LECTURE	MIT 8.821 Lecture 20 only through the GKPW/boundary-value and regulator setup. The local counterterms and renormalized two-point completion are used in Week 46. [LECTURES]
5	SOLVE	MIT 8.821 Problem Set 5, Problem 2(a–c): Poincare-patch geometry and scalar AdS propagators. [PROBLEMS]
6	OUTCOME	Derive the GKPW source prescription and identify precisely where a regulator enters before the on-shell bulk action can define a field-theory generating functional.

4.6 Week 46 — Scalar holographic renormalization and renormalized correlators

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 5.5.1, *Scalar*.
2	READ	Skenderis, *Lecture Notes on Holographic Renormalization*, through the scalar near-boundary expansion, counterterm construction and renormalized one-point prescription. [PAPER]
3	DERIVE	Solve the scalar expansion recursively; identify power and logarithmic divergences; construct local covariant counterterms; define S_{ren}; obtain $\langle O \rangle$ by varying the renormalized action and recover the nonlocal two-point function.
4	LECTURE	Return to the second part of MIT 8.821 Lecture 20 for local counterterms and the renormalized two-point function, now after the systematic scalar counterterm construction. [LECTURES]
5	SOLVE	Ammon–Erdmenger Exercise 5.5.1; then repeat the recursion at a value of Δ for which the denominator degenerates and a logarithmic term appears, identifying the corresponding scheme-dependent local term.
6	OUTCOME	Compute a boundary correlator from a finite variational problem; distinguish universal nonlocal data from local/contact counterterm dependence; and distinguish the near-boundary recursion from the separate interior-regularity condition that fixes the normalizable coefficient as a functional of the source.

4.7 Week 47 — Higher-point correlators and Witten diagrams

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 5.4.2 for the general Witten-diagram rules and the opening contact/exchange material of Sec. 6.2.1; reserve Sec. 6.1 for the protected precision tests of Week 49.
2	READ	Witten 1998, tree-level bulk-diagram discussion. [PAPER]
3	DERIVE	For a convergent, non-extremal toy scalar cubic interaction, solve perturbatively in the boundary source; differentiate the renormalized classical action three times and obtain the contact Witten diagram; write an exchange diagram using one bulk-to-bulk propagator and identify structurally the corresponding intermediate CFT operator/channel, without assuming a full conformal-block decomposition.
4	LECTURE	MIT 8.821 Lecture 21 only through the higher-point/Witten-diagram portion; stop before the lecture turns to Wilson loops, which begin in Week 51. [LECTURES]
5	SOLVE	MIT 8.821 Problem Set 5, Problem 2(d), then write and dimensionally check the AdS integral for one scalar three-point contact Witten diagram. [PROBLEMS]
6	OUTCOME	Translate bulk interaction vertices and propagators into boundary correlator structures and distinguish contact from exchange contributions.

4.8 Week 48 — Metric holographic renormalization, Fefferman–Graham expansion, and $\langle T_{\mu\nu} \rangle$

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 5.5.2, *Metric*; reserve Sec. 6.3 for the protected anomaly comparison of Week 49.
2	READ	de Haro–Solodukhin–Skenderis for the Fefferman–Graham expansion, gravitational counterterms and holographic stress tensor. [PAPER]
3	DERIVE	Starting from Einstein–Hilbert plus Gibbons–Hawking terms, put an asymptotically AdS metric in Fefferman–Graham form; solve Einstein's equations asymptotically to the first nontrivial coefficients, separating the locally determined coefficients from the state-dependent data carried by the undetermined part of $g_{(d)}$; translate curvature-sign conventions explicitly before importing formulas from de Haro–Solodukhin–Skenderis; add local covariant counterterms; vary the renormalized action with respect to the boundary metric to obtain $\langle T_{\mu\nu} \rangle$ and its trace anomaly in the adopted convention.
4	SOLVE	Ammon–Erdmenger Exercise 5.5.2: compute $g_{(2)}$ from Einstein's equation; then verify the conservation and trace Ward identities of the resulting renormalized stress tensor.
5	OUTCOME	Treat the boundary metric as the source for $T_{\mu\nu}$ and derive the anomaly from the logarithmic/divergent structure of the gravitational action.

4.9 Week 49 — Kaluza–Klein/operator matching and protected precision tests

Step	Action	Exact task and source
1	READ	Revisit Ammon–Erdmenger Secs. 5.3.1–5.3.3 in detail; use selected parts of Sec. 6.1 for the protected two-/three-point agreement and Sec. 6.3 for the conformal-anomaly test. For the anomaly comparison, read Henningson–Skenderis as the primary holographic calculation rather than relying on a textbook summary alone. [PAPER] For an open account of the mixed Kaluza–Klein modes, see D'Hoker–Freedman Sec. 8, especially the field-mixing and harmonic-expansion discussion. [PAPER]
2	DERIVE	Expand the coupled scalar fluctuation formed from the S^5 metric trace and RR four-form in scalar harmonics; for the physical half-BPS scalar tower with integer $k \geq 2$ derive $m^2L^2 = k(k-4)$, then $\Delta = k$ in standard quantization and the $\mathfrak{su}(4)_R$ representation $[0, k, 0]$. Check separately the BF-saturating $k = 2$ mode and explain why $k = 1$ does not furnish an independent scalar single-trace primary of $SU(N)$ SYM. Do not apply this mixed-field mass formula to an arbitrary ten-dimensional scalar harmonic.
3	SOLVE	Construct three explicit entries of the bulk/operator dictionary, including a scalar half-BPS mode and the metric/stress-tensor sector; then reproduce the leading N^2 scaling of the $\mathcal{N} = 4$ conformal-anomaly normalization.
4	OUTCOME	Explain which comparisons are protected against changes in λ and why their agreement does not imply that generic unprotected observables are coupling-independent.

4.10 Week 50 — Holographic RG flows and domain walls

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Secs. 9.1.1–9.1.2 and 9.2.1–9.2.2: relevant deformations, field-theory and holographic c-theorems, domain-wall flows and their fixed points.
2	READ	Skenderis Sec. 6 for holographic RG flows, source/vev deformations and Ward identities after completing the counterterm formalism of Week 46; do not treat these notes as a full Hamilton–Jacobi derivation. [PAPER]
3	DERIVE	Perturb a UV CFT by a relevant operator and map source/vev data to a bulk scalar; derive the domain-wall equations for $ds^2 = dr^2 + e^{2A(r)}\eta_{\mu\nu}dx^\mu dx^\nu$; identify AdS critical points with CFT fixed points; derive the monotonic holographic c-function under the stated energy condition.
4	SOLVE	Choose a scalar potential with two AdS extrema; linearize about both extrema and determine the corresponding operator dimensions; then determine whether a regular interpolating domain-wall solution exists rather than assuming that the two extrema are automatically connected, and analyze its UV-to-IR behavior if it does.
5	OUTCOME	Connect the Wilsonian language of Week 9 to radial bulk evolution without promoting the radial-coordinate/RG-scale correspondence beyond what the equations establish.

4.11 Week 51 — Wilson loops and fundamental-string minimal surfaces

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 5.6.
2	READ	Maldacena, *Wilson Loops in Large N Field Theories*. [PAPER]
3	DERIVE	Starting from the ordinary Wilson loop of Week 6, construct the scalar-coupled Maldacena–Wilson observable $W[C] = (1/N)\,\mathrm{Tr}\,\mathcal{P}\exp\oint_C ds\,[iA_\mu \dot{x}^\mu + \lvert\dot{x}\rvert\,\theta^I\phi_I]$; explain the local supersymmetry condition and reserve the term half-BPS for the standard special contours/choices that actually preserve half the supercharges. Relate the fundamental-representation loop to a fundamental-string worldsheet ending on the boundary contour; set up the Nambu–Goto surface for a rectangular loop; subtract the two straight-string quark masses and derive $V(L) = -\dfrac{4\pi^2}{\Gamma(1/4)^4}\dfrac{\sqrt{\lambda}}{L}$ in the convention $g^2_{\mathrm{YM}} = 4\pi g_s$.
4	LECTURE	MIT 8.821 Lecture 22 only through the Wilson-loop/Coulomb-potential portion; reserve its finite-temperature second half for Week 52. [LECTURES]
5	SOLVE	Ammon–Erdmenger Exercises 5.6.2–5.6.3: verify $\langle W\rangle = 1$ for the straight half-BPS line with the stated counterterm and calculate the circular Wilson loop holographically.
6	OUTCOME	Explain why the dual object is a fundamental string rather than a supergravity field, and separate the conformally fixed $1/L$ dependence from the dynamical $\sqrt{\lambda}$ coefficient.

4.12 Week 52 — Finite temperature: AdS black branes and gauge-theory thermodynamics

Step	Action	Exact task and source
1	READ	Selected Ammon–Erdmenger Secs. 11.1.1–11.1.4 for the thermal-QFT bridge, followed by Sec. 11.2.1 for planar black-brane thermodynamics; stop before the real-time Green-function material of Sec. 11.2.3.
2	DEEPEN	Natsuume Secs. 7.1–7.2 for an independent calculation of the planar AdS black brane and its thermodynamic quantities.
3	DERIVE	Write the planar AdS_5 black-brane geometry; obtain the Hawking temperature from Euclidean regularity or surface gravity; use the Week-43 normalization $G_5 = \pi L^3/(2N^2)$ in the Bekenstein–Hawking area law; derive $s = \pi^2 N^2 T^3/2$ in the leading large-N, large-λ, two-derivative classical-supergravity regime.
4	LECTURE	Use the finite-temperature second half of MIT 8.821 Lecture 22 for the planar black-brane generalization. Lecture 23 is optional deepen material for global-AdS/Hawking–Page and chemical-potential physics rather than the core planar derivation. [LECTURES]
5	SOLVE	MIT 8.821 Problem Set 5, Problem 3: finite-temperature color screening using the Wilson-loop/string configuration. [PROBLEMS]
6	OUTCOME	Derive the thermal dictionary from horizon geometry and keep equilibrium thermodynamics distinct from real-time response.

4.13 Week 53 — Lorentzian holography, retarded functions, Kubo formulae, and transport

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Sec. 11.1.5 and Sec. 11.2.3 for real-time thermal field theory and holographic Green functions, followed by selected Secs. 12.1 and 12.3, especially Secs. 12.3.3 and 12.3.5 for the shear Kubo relation and holographic viscosity.
2	DEEPEN	Use selected Natsuume Ch. 9 for linear response/Kubo/viscous hydrodynamics and Sec. 10.4 for the Lorentzian prescription; omit the earlier scalar-dictionary material already developed in Weeks 44–46.
3	READ	Son–Starinets, *Minkowski-Space Correlators in AdS/CFT Correspondence*, together with Kovtun–Son–Starinets for the viscosity/entropy result and its original scope. [PAPER] [PAPER]
4	DERIVE	Using the Fourier convention $e^{-i\omega t}$, solve a linear bulk fluctuation with infalling horizon condition; evaluate the renormalized radial canonical momentum/boundary flux and derive the retarded Green function. For the isotropic shear channel, derive $\eta = -\lim_{\omega\to 0} \operatorname{Im} G^R_{T_{xy}T_{xy}}(\omega, \mathbf{0})/\omega$ and recover $\eta/s = 1/(4\pi)$ in the two-derivative Einstein/classical-supergravity limit.
5	STATUS	Keep the historical status precise: Son–Starinets presents the retarded rule as a prescription/conjecture and tests it nontrivially; do not retroactively attribute the later full Schwinger–Keldysh gravitational construction to that paper. Likewise, treat $\eta/s = 1/(4\pi)$ here as the isotropic two-derivative Einstein/classical-supergravity result, not as an exact statement of all holographic theories.
6	SOLVE	For the T_{xy} channel with the stated $e^{-i\omega t}$ convention, write both independent near-horizon solutions, select the infalling one, propagate its source/response data to the boundary, and verify the corresponding dissipative sign of $\operatorname{Im} G_R(\omega, \mathbf{0})$.
7	OUTCOME	Given a real-time bulk fluctuation problem, state the correct horizon condition, boundary source, renormalized response and retarded correlator without appealing to Euclidean regularity alone.

4.14 Week 54 — Holographic entanglement entropy and synthesis

Step	Action	Exact task and source
1	READ	Ammon–Erdmenger Secs. 15.6.1–15.6.2: field-theory entanglement entropy and its holographic realization.
2	READ	Ryu–Takayanagi for the original static proposal, Headrick–Takayanagi for the modern homology condition, and the extended Ryu–Takayanagi treatment Secs. 7.1.1–7.2 for the complete higher-dimensional strip construction. Use Sec. 6.3 only if the AdS_3 interval is needed as a warm-up. [PAPER] [PAPER] [PAPER]
3	DERIVE	Starting from $\rho_A = \mathrm{Tr}_{\bar{A}}\,\rho$ and $S_A = -\,\mathrm{Tr}\,\rho_A \ln \rho_A$, formulate the static RT surface as codimension two, anchored on ∂A, and homologous to A, distinguishing historical proposal from later homology formulation. CORE: for a width-ℓ strip in Poincare AdS_5 at fixed time and transverse area V_2, derive its area functional $\mathcal{A} = L^3 V_2 \int_{-\ell/2}^{\ell/2} dx\, z^{-3}\sqrt{1+(z')^2}$, obtain the first integral and turning-point relation $\ell \propto z_*$, and extract $S_A = \mathcal{A}/(4G_5)$ with the leading V_2/ϵ^2 divergence and finite $-V_2/\ell^2$ scaling. DEEPEN: evaluate the gamma-function coefficient and compare the AdS_3 interval calculation.
4	LECTURE	MIT 8.821 Lecture 24: entanglement entropy. [LECTURES]
5	SOLVE	Construct a one-page derivational dictionary linking: at fixed λ, $g_s = \lambda/(4\pi N)$ and each additional closed-string handle to a factor $g_s^2 \sim N^{-2}$; $\alpha'/L^2 = \lambda^{-1/2}$ to the string-scale/curvature expansion parameter; in standard quantization, boundary sources to the leading/non-normalizable datum and vevs to the normalizable response; radial evolution to RG scale with the flow/scheme caveat of Week 50; temperature to horizons; and static entanglement to minimal area.
6	OUTCOME	Reconstruct the central calculations of Weeks 41–54 in their correct approximation regimes and label each statement as exact dictionary input, large-N result, strong-coupling result, semiclassical gravity result or protected comparison.

5 References and online resources

Reading access and equivalence. The free Tong QFT/string/gauge/SUSY lectures, MIT QFT courses, and the open CFT and holography notes support substantial parts of the route. Polchinski I and Weinberg are book references rather than presumed freely available assignments. In particular, Tong's string lectures focus on the bosonic string: their short discussions of superstrings do *not* replace the complete RNS/GSO and Type-II treatment assigned for Weeks 27–30 from Polchinski II. Before those four weeks, confirm learner access to the indicated material or qualify another *complete* source for those specific derivations; a summary of the Type-II spectrum is not an adequate substitution. For the later D3/dictionary and protected tests, D'Hoker–Freedman Secs. 3–6 and 8, MIT 8.821 Lectures 16–24, and the original papers provide open alternatives to the relevant Ammon–Erdmenger readings. [PAPER] [PDF]

5.1 Quantum field theory

Resource	Use in the program and access
Weinberg, *The Quantum Theory of Fields*, Vols. I–II	Relativistic fields, S-matrix, path integrals, Yang–Mills, BRST, renormalization, RG, OPE, and anomalies.
David Tong, QFT	Working treatment of classical/free/interacting fields and fermions; problem sheets used in Weeks 1–5. [PDF] [COURSE] [PROBLEMS] [PROBLEMS] [PROBLEMS] [PROBLEMS]
David Tong, Gauge Theory	Wilson operators, running couplings, anomalies, and large-N double-line counting. [PDF] Large N: [PDF]
MIT 8.323, QFT I	Lectures supporting Weeks 1–5. [COURSE] [LECTURES]
MIT 8.324, QFT II	Gauge theory, renormalization and RG lectures/problem sets used in Weeks 6–10. [LECTURES] [PROBLEMS]

5.2 String theory

Resource	Use in the program and access
Polchinski, *String Theory*, Vol. I	Bosonic string quantization, worldsheet CFT, ghosts/BRST, amplitudes, background fields, compactification, T-duality, and D-branes.
Polchinski, *String Theory*, Vol. II	RNS superstrings, Type-II spectra, low-energy supergravity, RR charge, and D-branes.
David Tong, String Theory	Parallel working treatment and problem sheets for Weeks 14–26. [PDF] Sheets 1–4: [PROBLEMS] [PROBLEMS] [PROBLEMS] [PROBLEMS]

5.3 Conformal field theory and supersymmetry

Resource	Use in the program and access
Rychkov, *EPFL Lectures on Conformal Field Theory in $D \geq 3$*	Conformal kinematics, radial quantization, unitarity, and the OPE. [PDF]
Ammon–Erdmenger, *Gauge/Gravity Duality*, Ch. 3	Four-dimensional conformal symmetry, $\mathcal{N} = 4$ SYM, superconformal symmetry, and protected operators used in Weeks 31–40.
Simmons–Duffin, *TASI Lectures on the Conformal Bootstrap*	Independent treatment of stress tensors, representations, radial quantization, reflection positivity, and OPE structure. [PDF]
Weinberg, *The Quantum Theory of Fields*, Vol. III	Supersymmetry algebra, superfields, supersymmetric gauge theory, and extended supersymmetry.
David Tong, Supersymmetric Field Theory	Working treatment of supersymmetry, superspace, and supersymmetric gauge theory; problem sheets used in Weeks 34–36. [PDF] [PROBLEMS] [PROBLEMS]

5.4 AdS/CFT and holography

Resource	Use in the program and access
Ammon–Erdmenger, *Gauge/Gravity Duality*, Chs. 4–15 selectively	Main text for Type-II/D-brane prerequisites, the D3 construction, the AdS/CFT dictionary, correlators, holographic renormalization, RG flows, Wilson loops, thermal physics, and entanglement.
Natsuume, *AdS/CFT Duality User Guide*	Finite-temperature holography, Lorentzian correlators, linear response, hydrodynamics, and transport.
MIT 8.821, Hong Liu	Lectures and problems used from large N through the canonical holographic observables. [LECTURES] [PROBLEMS]
Maldacena 1998	D3-brane decoupling limit and the original AdS/CFT conjecture. [PAPER]
Gubser–Klebanov–Polyakov	Generating-functional prescription. [PAPER]
Witten, *Anti-de Sitter Space and Holography*	Field–operator map, generating functionals, and Witten diagrams. [PAPER]

Resource	Use in the program and access
Breitenlohner–Freedman, *Stability in Gauged Extended Supergravity*	Original positive-energy/boundary-condition stability analysis underlying the BF mass threshold; not replaceable by the indicial-root calculation alone. [PAPER]
Klebanov–Witten, *AdS/CFT Correspondence and Symmetry Breaking*	Primary reference for the two admissible scalar quantizations in the Breitenlohner–Freedman window. [PAPER]
Skenderis, *Lecture Notes on Holographic Renormalization*	Scalar/gravitational holographic renormalization and holographic RG. [PAPER]
de Haro–Solodukhin–Skenderis	Fefferman–Graham expansion and holographic stress tensor. [PAPER]
Henningson–Skenderis, *The Holographic Weyl Anomaly*	Primary holographic calculation of the conformal anomaly, including the $d = 4$ $\mathcal{N} = 4$ SYM comparison used in the protected precision tests. [PAPER]
Maldacena, Wilson loops	Fundamental-string prescription for Wilson loops. [PAPER]
Son–Starinets	Lorentzian prescription for retarded correlators. [PAPER]
Kovtun–Son–Starinets	Shear-viscosity to entropy-density ratio in the two-derivative gravity regime and the historical universality discussion. [PAPER]
Brown–Henneaux	Asymptotic-symmetry central charge $c = 3L/(2G_3)$ used in the AdS_3/CFT_2 interval check. [PAPER]
Headrick–Takayanagi	Primary geometric formulation used here to support the standard homology requirement and its entropy-inequality role. [PAPER]
Ryu–Takayanagi	Static holographic entanglement entropy. [PAPER]
Ryu–Takayanagi, *Aspects of Holographic Entanglement Entropy*	Detailed minimal-curve and higher-dimensional strip calculations; Secs. 6.3 and 7.1.1–7.2. [PAPER]